

Moral Hazard Economies

Outline

- ① Competitive Capital Markets: the role of Covenants
- ② Competitive Screening under Moral Hazard:
Parlour-Rajan (2001), Bisin-Guaitoli (2004)

Motivation

Analyze the relationship between incentives and competition under moral hazard

We focus on capital markets

We model the capital market as a setting where several investors strategically compete over contracts to finance a representative entrepreneur

Moral Hazard in Financial Markets

Owner-Manager: The manager's actions affect the returns of the project. The owner cannot use a compensation scheme that depends on the actions of the manager

Insurance: The insuree's actions affect the probability of accidents

Strategic Default: Debtors may divert resources to themselves at the expense of investors

Exclusive vs Non-Exclusive Competition

Contractual externalities arise: the entrepreneur's decisions depend on the entire profile of trades she enters into

Can this externality be controlled by letting investors free to design sophisticated contracts?

Most economic applications impose **de-jure exclusivity**: agents are restricted to trade at most one contract (the contractual externality is minimized)

At the other extreme, applications consider **non exclusivity**: a competitor cannot write contracts contingent on the offers that agents receive from his rivals (no instrument to limit the contractual externality)

Exclusive vs Non-Exclusive Competition in Capital Markets

No legal system imposes that entrepreneurs must raise funds from a single investor

Financial transactions are not fully secret and investors can use the available information to design financial covenants to control firms' financing policy

Related Literature

- **Non-Exclusive Competition in Financial Markets under Moral Hazard:** Hellwig (1983), Arnott-Stiglitz (1990), Bizer-De Marzo (1992), Parlour-Rajan (2001), Bisin-Guaitoli (2004), Bisin-Rampini (2006), Khalil-Martimort-Parigi (2007), Attar-Chassagnon (2009), Bennardo-Pagano-Piccolo (2010)
- **Common agency:** Martimort-Stole (2002), Peters (2001), Segal-Whinston (2003)
- **Debt Covenants:** Smith-Warner (1979), Gorton-Khan (2000), Garleanu-Zwiebel (2009)

The Economy: Building on Holmstrom-Tirole (1997, 1998)

We study a production economy populated by a single (representative) entrepreneur and $N \geq 2$ competing investors

The entrepreneur takes a non-contractible effort $e = \{L, H\}$ with $H > L$

Given the investment $I \in \mathbb{R}_+$, we have:

$$\begin{cases} GI & \text{with probability } \pi_e & \text{(good state)} \\ 0 & \text{with probability } 1 - \pi_e & \text{(bad state)} \end{cases}$$

with $G \in \mathbb{R}_+$, and $\Delta\pi = \pi_H - \pi_L > 0$

Payoffs (I)

The entrepreneur is risk-neutral and protected by limited liability.

For any amount $A \in \mathbb{R}_+$ of assets in place, if she gets an aggregate investment I in exchange for a **promised** aggregate repayment R :

$$U(I, R, e) = \begin{cases} \pi_H(G(I + A) - R) - A & \text{if } e = H \\ \pi_L \max\{(G(I + A) - R), 0\} + B(I + A) - A & \text{if } e = L \end{cases}$$

If $G(I + A) < R$, strategic default takes place: investors seize the project's cash flow. The entrepreneur chooses $e = L$ and gets the private benefit $B(I + A)$

Payoffs (II)

If default does not take place, the payoff to investor i is

$$V^i(I_i, R_i, e) = \pi_e R_i - I_i \quad \text{with } e \in \{L, H\},$$

If default takes place, each investor i is repaid according to a **pro-rata** rule

$$R_i = G(I + A) \frac{I_i}{I}$$

Competition

Investors compete over (menus of) financial contracts:

- ① Each investor i proposes a menu of financial contracts $C^i = (I_i, R_i(\cdot))$, where $I_i \in \mathbb{R}_+$ is an investment and $R_i(\cdot) : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a schedule of repayments contingent on the aggregate investment (external finance)
- ② After observing the array of offered menus, the entrepreneur takes her portfolio and effort decisions, and $C = (I, R)$ is the corresponding aggregate allocation

Strategies and Equilibrium

A pure strategy for the entrepreneur is a mapping associating to each profile of offers $(C^1, C^2, \dots, C^N)$ an element of $\{L, H\} \times \{0, 1\}^N$

The equilibrium concept is pure strategy Subgame Perfect Equilibrium (SPE)

Feasible Allocations (1)

$$\mathcal{H} = \{(I, R) : U(I, R, H) \geq U(I, R, L)\} \quad \mathcal{L} = \{(I, R) : U(I, R, L) \geq U(I, R, H)\}$$

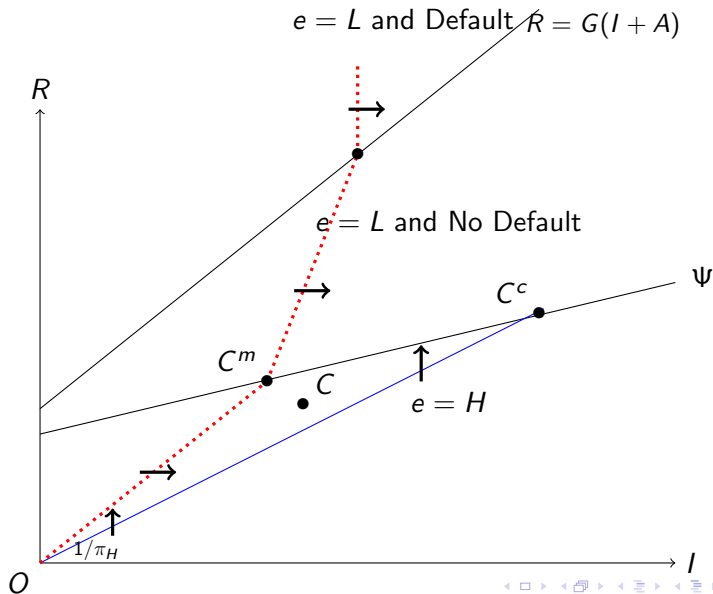
The frontier of $\mathcal{H}$ is: $\Psi = \{(I, R) : U(I, R, H) = U(I, R, L)\}$

Following Holmstrom-Tirole (1997, 1998), one has:

$$\pi_H G > 1 > \pi_L G + B \tag{1}$$

$$G - \frac{B}{\Delta\pi} < \frac{1}{\pi_H} \tag{2}$$

Feasible Allocations (2)

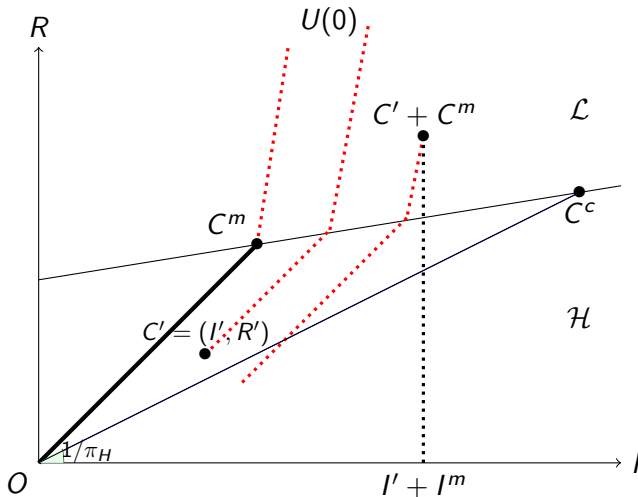


Bilateral Contracts: Non-Exclusive Competition

A contract $C^i = (I_i, R_i) \in \mathbb{R}_+^2$

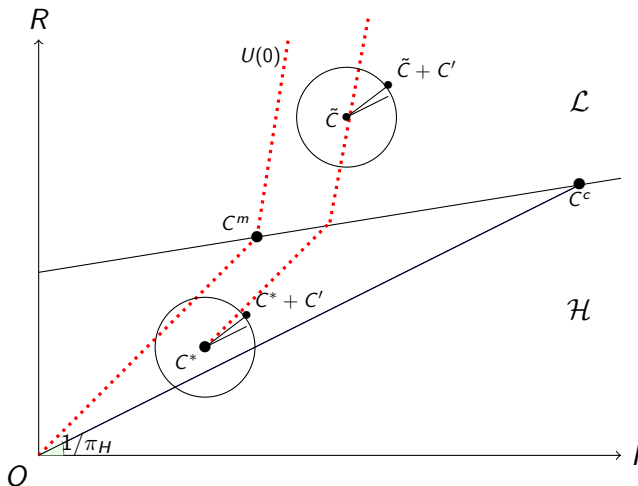
Lemma 1: There is an equilibrium such that one of the investors proposes $C^m = (I^m, R^m)$, and all the others propose the null contract $(0, 0)$. At equilibrium, the entrepreneur accepts the offer (I^m, R^m) and selects $e = H$

Monopoly Equilibrium: only $C^m = (I^m, R^m)$ is Offered



Equilibrium Analysis: $C^* \notin \Psi$

There is a $\tilde{C} \in L$: $U(C^*, H) = U(\tilde{C}, L) \implies \exists$ **inactive investors**



Equilibrium Analysis: $C^* \notin \Psi$ (2)

- ① **No deviation for inactive investors:** $\tilde{C}$ must remain available if any of the investors withdraws his offer
- ② It follows that no contract at a price $p \in (G, G + \frac{B}{\pi_L})$ is proposed
- ③ The price of any offered contract offered is in the set $\{G, G + \frac{B}{\pi_L}\}$

Bilateral Contracts: Constrained Efficiency

Proposition 1: (Attar-Casamatta-Chassagnon-Decamps (2019))

If $C^ = (I, R)$ is an equilibrium allocation, then $C^* \in \Psi$*

Intuition: If not, any inactive lender who offers $\bar{C} - C^*$ can deviate removing his offer and proposing a small contract that induces the entrepreneur to choose $e = H$

The result obtains since no investor is essential to provide the entrepreneur her equilibrium payoff

Menus vs Take-It or Leave-It Offers

Each unilateral deviation towards a menu can be reconstructed as a deviation in take-it or leave-it offers

On the equilibrium path, menus in principle allow for a richer structure of threats

However, we show that such threats do not prevent unilateral deviations inducing $e = H$

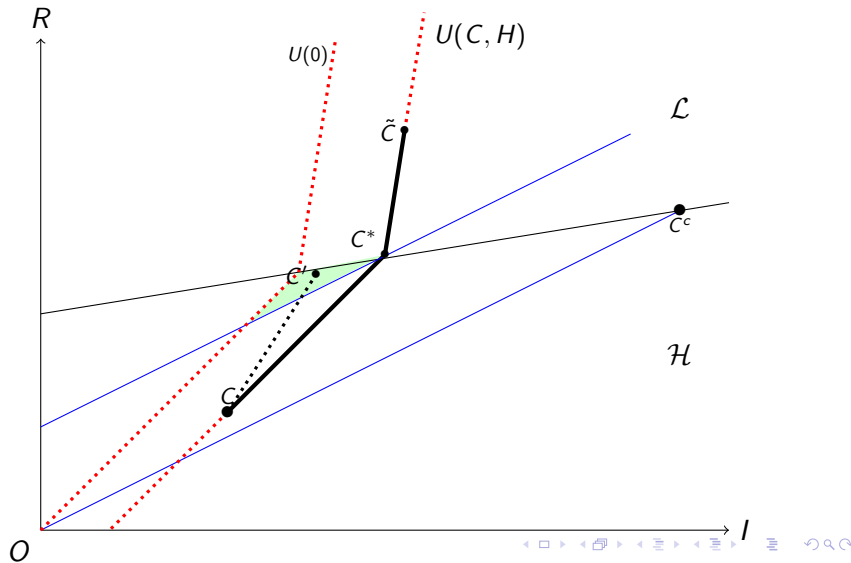
Equilibrium Characterization

Proposition 2:

- ① **Mild Moral Hazard** ($B \leq \pi_H G - 1$): *every allocation $(I, R) \in \Psi$ can be supported at equilibrium*
- ② **Strong Moral Hazard** ($B > \pi_H G - 1$): *only the allocation (I^m, R^m) can be supported at equilibrium*

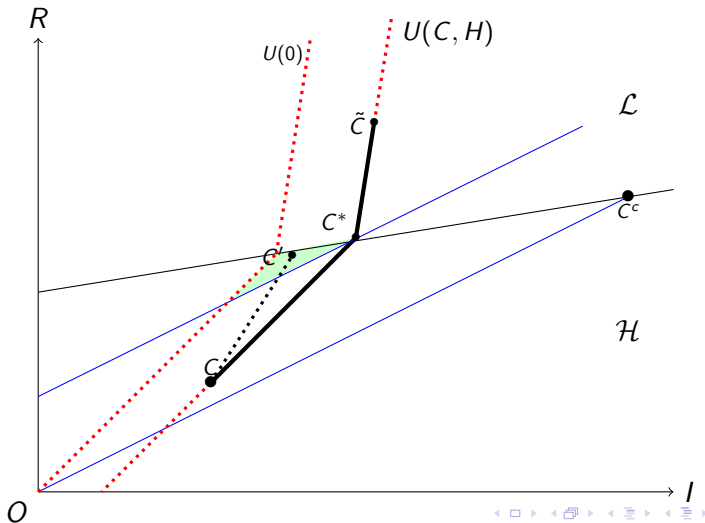
Strong Moral Hazard: only (I^m, R^m) at Equilibrium

A deviation inducing C' is profitable



Mild Moral Hazard: any $(I, R) \in \Psi$ at Equilibrium

B is small: any deviation inducing C' is such that the entrepreneur defaults



Financial Covenants

Contracts can now be contingent on the entrepreneur's external aggregate investment

Each investor gets additional instruments to discipline the entrepreneur whenever she departs from some target aggregate results

The set of available punishments depends on the information available to investors

Covenant violation *after* investment

Suppose that outside financing is observed only after production takes place

This enables investors to require higher interest rates when undesired levels of cash flows are observed

Any investor can set punishments to alleviate the entrepreneur's threat to overborrow: raise in interest rates to eliminate the entrepreneur's rent when she chooses $e = L$ and repays all investors

Equilibrium Indeterminacy (I)

If repayments R_i can be contingent on the total investment $(I + A)$, market equilibria are indeterminate and Pareto-ranked

Proposition 3 (ACCD, 2019):

For each $N \geq 2$ there exists a $U_N < U(I^c, R^c, H)$ such that any (I^, R^*) satisfying*

$$U(I^*, R^*, H) \geq U_N$$

is sustained at equilibrium.

If moral hazard is strong ($B > \pi_H G - 1$), $\lim_{N \rightarrow \infty} U_N = U(0)$.

Equilibrium Indeterminacy (II)

Each investor $i = 1, \dots, N$ posts $M^* = \{(0, 0); (\frac{I^*}{N}, R^*(.)); (\frac{\hat{I}}{N}, \hat{R}(.))\}$:

$$R^*(I) = \begin{cases} \frac{R^*}{N} & \text{if } I = I^*, \\ R^* - \frac{N-1}{N}GI^* & \text{if } I = \frac{I^*}{N}, \\ R^*(I) = G(I+A) & \text{if } I \notin \{I^*, \frac{I^*}{N}\} \end{cases}$$

The investment level $\hat{I}$ is such that

$$U(I^*, R^*, H) = B(\hat{I} + A) - A, \quad (3)$$

$\Rightarrow \hat{I} > I^* \geq 0$. Also, $\hat{R}(I) = G(I+A) \forall I \in \mathbb{R}_+$.

Equilibrium Indeterminacy (III): Equilibrium Strategies

At equilibrium, the entrepreneur is indifferent between

- Trading $\left(\frac{I^*}{N}, \frac{R^*}{N}\right)$ with each investor, and choosing $e = H$
- Trading $\left(\frac{\hat{I}}{N}, \frac{\hat{R}}{N}\right)$ with each investor, and defaulting

Relationship with Bilateral Contracting (1)

- 1 No investor can gain by offering an additional investment on top of what is offered by his rivals (given the covenants issued on the equilibrium path)
- 2 Each investor can potentially gain by serving the entire market alone (using an extreme form of covenants at the deviation stage)

Relationship with Bilateral Contracting (2)

Given the equilibrium covenants:

- ① Investors cannot profitably deviate by decreasing the entrepreneur's utility and inducing $e = H$ (**No investor is indispensable to provide the entrepreneur with her equilibrium payoff**)
- ② No investor can profitably deviate by increasing the entrepreneur's utility and inducing $e = H$ (**U_N is chosen to guarantee that every such deviation induces the entrepreneur to strategically default**)

Covenant Violation *Before* Investment

Investors also observe outside financing before production takes place

This enlarges the set of available financial covenants: investors can adjust their contractual terms according to the entrepreneur's intended financial policy

- 1 They can include loan acceleration clauses if they observe excessive debt issuance
- 2 They can propose additional financing if they observe insufficient borrowing: issuing a contingent line of credit

Loan Accelerations and Lines of Credit

A contract for investor i as $(l_i, R_i(\cdot), l_i^+(\cdot), R_i^+(\cdot))$ with $l_i > 0$, $l_i^+(l_0) \geq -l_i$

$R_i(\cdot)$ and $R_i^+(\cdot)$ are functions of the initial debt $l_0 \equiv \sum_i l_i$ and of the final outside financing $I \equiv l_0 + I^+(l_0)$

- ① A loan acceleration clause following an undesired level of debt l_0 :
 $l_i^+(l_0) = -l_i$ and $R_i^+(l_0, I) = -R_i(l_0, I) \forall I$
- ② A line of credit : $l_i^+(l_0) > 0$, and $R_i^+(l_0, I) = p l_i^+(l_0)$, with $l_i^+(l_0)$ being the credit limit available at a unit price $p \equiv \frac{R_i^+(l_0)}{l_i^+(l_0)}$

Indeterminacy Again

Proposition 4 (AACD, 2019)

Proposition 3 extends to the case in which investors can write covenants contingent on the initial debt l_0 .

Intuition: If initial debt is observable, an investor could in principle gain by offering a more competitive loan while protecting himself with a loan acceleration clause: he can therefore try to enforce exclusivity by asking to be reimbursed if he observes additional debt issuance.

Such a deviation is however blocked by investors' ability to offer lines of credit contingent on the initial debt.

Competitive Screening under Moral Hazard

Motivation

Study the relationship between incentives and competition in insurance markets affected by Moral Hazard.

We think of the insurance market as a setting where several intermediaries compete over the contracts they offer to a single representative agent (consumer-borrower).

Insurance Markets

Several insurance markets seem not to operate under exclusive assumptions: both life insurance and annuity contracts are typically non-exclusive.

Competition in non-exclusive markets can hardly be reconciled with competitive outcomes.

Annuity market (UK). “There is strong evidence that annuities are priced higher than what is suggested by actuarial considerations, despite the existence of a fairly high number of companies” (British Survey of Annuity Pricing, 2006).

Positive results

- ① A simple characterization of (pure strategy) equilibria in financial markets where intermediaries compete in the presence of moral hazard on the agents' side.
- ② Non-exclusivity induces lower provision of insurance (credit) and distorts the distribution of surplus relative to the exclusive scenario.
- ③ Developing examples to emphasize that positive profits and nonlinear prices are a typical feature of non-exclusive markets.

Normative Results

- 1 Identify situations where (pure strategy) equilibria fail to be second best efficient.
- 2 Optimality can be recovered if the social planner can enforce exclusivity clauses.
- 3 Consider the situation where the planner cannot directly enforce exclusivity: it is impossible to prevent intermediaries from side contracting: a form of public insurance is anyhow welfare improving.

A General Set-Up

A standard two-period, two-state insurance economy with by a single agent and $n \geq 2$ intermediaries.

The agent takes a non-contractible effort $e = \{a, b\}$ with $a > b$.

The choice of e affects the probability distribution of the agent's wealth in the second period: $W = (w_1, w_2) \in \mathbb{R}_+^2$, with $w_2 > w_1$ and:

$$\begin{cases} w_2 & \text{with probability } \pi_e & \text{(good state)} \\ w_1 & \text{with probability } 1 - \pi_e & \text{(bad state)} \end{cases}$$

with $\pi_a > \pi_b$. The cost of effort is linear: $C(e) = e$, and $\Delta = a - b$.

Competition

Intermediaries compete over insurance contracts.

An insurance contract is a pair of state-contingent transfers from intermediaries to agents: $d^i = (d_1^i, d_2^i) \in \mathbb{R}^2$

- If $d_1^i > 0$ and $d_2^i < 0$, the contract d^i is with positive insurance.
- If $d_1^i < 0$ and $d_2^i > 0$, the contract d^i is with negative insurance.
- We let $p^i = \left| \frac{d_2^i}{d_1^i} \right|$ be the implicit price of this contract.
- The consumer can choose any fraction $\lambda_i \in [0, 1]$ of d^i .

Payoffs

The VNM utility $u(c_s) : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is strictly concave.

Agent's consumption in state $s \in \{1, 2\}$ is: $c_s = w_s + \sum_{i \in J} \lambda_i d^i$.

$$\tilde{U}(c_1, c_2, e) = (1 - \pi_e)u(c_1) + \pi_e u(c_2) - e,$$

$$U(c_1, c_2) = \tilde{U}(c_1, c_2, e(c_1, c_2)), \quad \text{with} \quad e(.) \in \arg \max_e \tilde{U}(c_1, c_2, e).$$

$$V^i(d^i, e, \lambda_i) = -((1 - \pi_e)d_1^i + \pi_e d_2^i)\lambda_i = -(\vec{\pi}_e \cdot d^i)\lambda_i, \quad \text{for } i = 1, 2, \dots, J$$

Strategies and Timing

A contract d^i cannot be made contingent on other intermediaries' offers and on agents' choices either:

- Intermediaries simultaneously post their offers $(d^1, d^2, \dots, d^J)$.
- The agent takes her trade and effort decisions: for every given offers' array, she selects an element in $\{a, b\} \times [0, 1]^J$.

Equilibrium

We consider pure strategy Subgame Perfect Equilibria:

- The agent optimally chooses her effort level together with the desired composition of her portfolio. That is, for every offered array $D = (d^1, d^2, \dots, d^J)$:

$$((\lambda_i(D))_{i \in J}, e(D)) \in \arg \max_{e, \lambda} \pi_e u(c_2) + (1 - \pi_e) u(c_1) - e \quad (4)$$

- For every array $D^{-i} = (d^1, \dots, d^{i-1}, d^{i+1}, \dots, d^J)$ of competitors' offers, every intermediary i chooses d^i so to maximize V^i subject to (4).

Incentive Compatibility

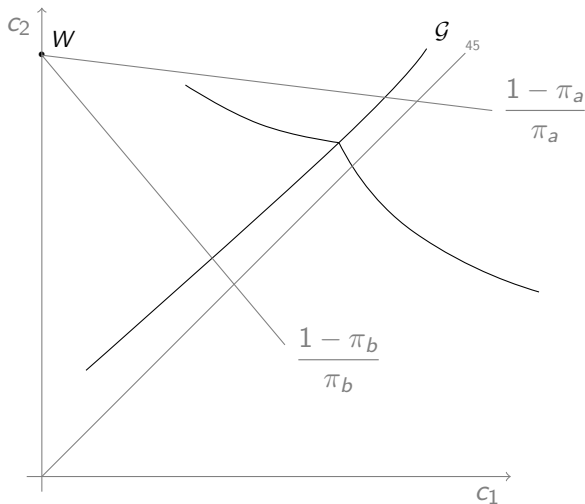
$\mathcal{A} = \{(c_1, c_2) \in \mathbb{R}_+^2 : \tilde{U}(c_1, c_2, a) \geq \tilde{U}(c_1, c_2, b)\}$ and $\mathcal{B}$ its complement.

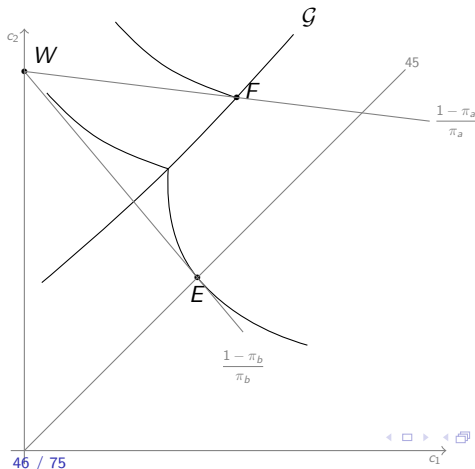
The frontier of $\mathcal{A}$ (incentive frontier) is

$$\mathcal{G} = \{(c_1, c_2) \in \mathbb{R}_+^2 : \tilde{U}(c_1, c_2, a) = \tilde{U}(c_1, c_2, b)\}.$$

The Consumption Set

The $\mathcal{G}$ frontier is $(\pi_a - \pi_b)[u(c_2) - u(c_1)] = \Delta$.



$$\begin{cases} \text{If } \Delta > \tilde{\Delta}, & e = b \text{ is chosen and } E \text{ is selected} \\ \text{If } \Delta \leq \tilde{\Delta}, & e = a \text{ is chosen and } F \text{ is selected} \end{cases}$$


Non-Exclusivity: Consumer's Behavior

For every given $(d^1, d^2, \dots, d^J)$, the set of available consumptions:

$\mathcal{F} = \{(c_1, c_2) \in \mathbb{R}_+^2 : (c_1, c_2) = W + \sum_{i \in J} \lambda_i d^i\}$ is convex.

Take any optimal choice $C = (c_1, c_2) \in \mathcal{F}$ and the corresponding

$$\tau^C = \frac{1 - \pi_e}{\pi_e} \frac{u'(c_1)}{u'(c_2)}.$$

- If the offer of intermediary i is bought ($\lambda_i \in (0, 1]$), then necessarily $p^i \leq \tau^C$.
- In particular, all contracts of price $p^i < \tau^C$ will be entirely bought ($\lambda_i = 1$).

Low Effort Equilibria (Δ High)

Proposition (Bisin-Guaitoli, 2004):

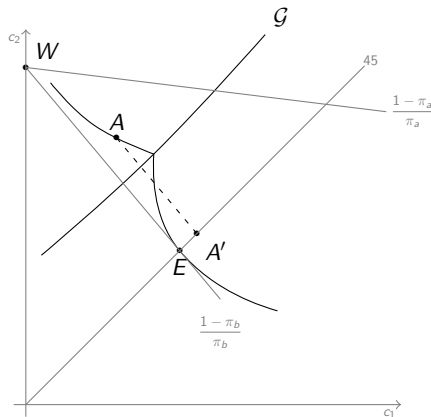
If

$$(1-\pi_a)u(c_1)+\pi_a u(c_2)-\Delta < u((1-\pi_b)c_1+\pi_b c_2) \quad \forall (c_1, c_2) \in \mathcal{A}, \quad (5)$$

then the unique equilibrium allocation involves zero-profit, full insurance and low effort.

Interpreting (5): The agent strictly prefers low effort and full insurance than high effort and partial insurance.

Non-Exclusivity and Low Effort Equilibria (Δ High)



At least two intermediaries offer a contract delivering the competitive allocation E . No intermediary can profitably deviate inducing high effort.

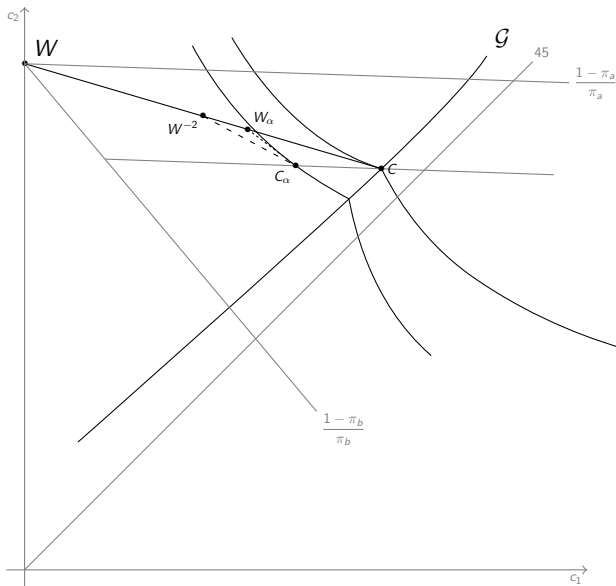
High Effort Equilibria on the Incentive Frontier $\mathcal{G}$

Proposition (Bisin-Guaitoli, 2004):

If an equilibrium allocation $C = (c_1, c_2) \in \mathcal{A}$ belongs to the incentive frontier $\mathcal{G}$, then:

- ① *All intermediaries are fully active:*
 $\lambda_1 = \lambda_2 = \dots = \lambda_J = 1 = 1.$
- ② *All intermediaries charge the same price:*
 $p^1 = p^2 = \dots = p^J = \tau^C$ *and each of them earns a strictly positive profit.*

Non-Existence of Equilibria on the Frontier $\mathcal{G}$

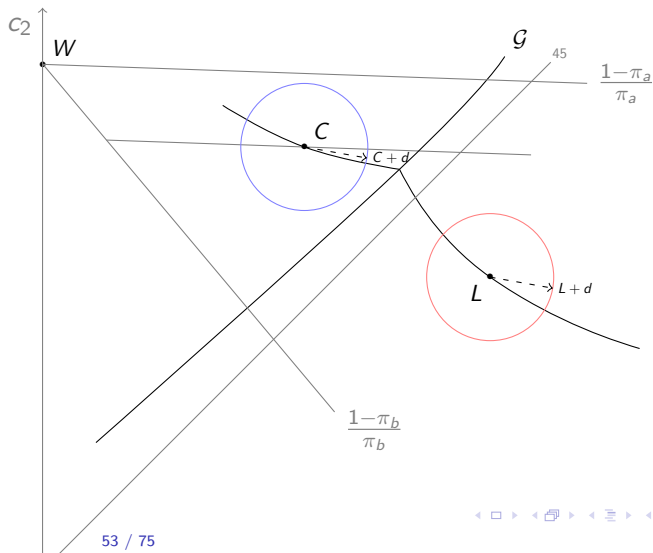


High Effort Equilibria which are NOT on $\mathcal{G}$

- Latent allocations are needed.
- Necessary conditions for equilibrium.
- Examples.

The need for latent allocations

For $C \in \mathcal{A}$ to be supported at equilibrium, there should be an $L \in \mathcal{B}$:
 $U(C) = U(L)$.



Characterizing the latent allocation L

- No profitable deviation for latent intermediaries. The allocation L must remain available if one latent intermediary removes his offer:

$$\tau^L c_1 + c_2 = \tau^L l_1 + l_2 \quad \text{with} \quad \tau^L = \frac{(1 - \pi_b)u'(c_1)}{\pi_b u'(c_2)}. \quad (6)$$

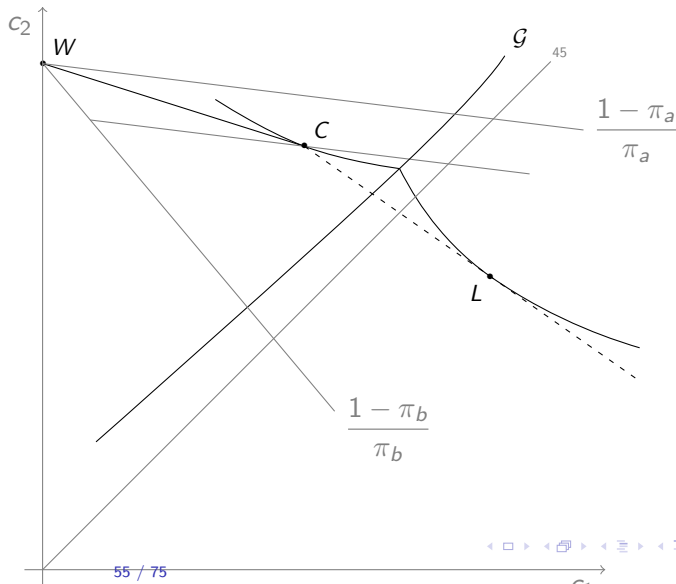
- No profitable deviation for active intermediaries. It is not possible to gain by reducing the agent's utility:

$$\pi_a(1 - \pi_a)(u'(c_1) - u'(c_2)) = \pi_a(1 - \pi_b)u'(l_1) - \pi_b(1 - \pi_a)u'(l_2) \quad (7)$$

which corresponds to:

$$(\nabla U_L - \nabla U_C) \text{ is collinear to } \vec{\pi}_a = ((1 - \pi_a), \pi_a)$$

Equilibrium $C \notin \mathcal{G}$: One Active and Several Latent Intermediaries



Example 1: One Active Intermediary

We take: $u(c) = c^\gamma$, $\gamma \in (0, 1)$ (CRRA).

We construct an equilibrium with only one active intermediary.

- Intermediary 1 offers the contract from W to C ($d^1 = C - W$),
- Intermediary 2 and 3 offer the contract from C to L ($d^2 = d^3 = L - C$),
- All other intermediaries do not make any offer $d^4 = d^5 = \dots = d^J = (0, 0)$,
- The agent fully accepts the offer of intermediary 1, rejects all the others and chooses $e = a$.

Necessary conditions:

$$U(C) = U(L) \quad (8)$$

$$\tau^L c_1 + c_2 = \tau^L h_1 + l_2 \quad (9)$$

$$\pi_a(1 - \pi_a)(u'(c_1) - u'(c_2)) = \pi_a(1 - \pi_b)u'(h_1) - \pi_b(1 - \pi_a)u'(l_2) \quad (10)$$

Sufficient conditions:

$$\frac{(1 - \pi_b)}{\pi_b} \geq \tau^L > \tau^C > \frac{w_2 - c_2}{c_1 - w_1} > \frac{(1 - \pi_a)}{\pi_a} \quad (11)$$

$$w_1 + 2(h_1 - c_1) = w_2 + 2(l_2 - c_2) \quad (12)$$

$$\frac{w_2 + 3(l_2 - c_2)}{w_1 + 3(h_1 - c_1)} < \frac{l_2}{h_1}, \quad -\vec{\pi}_a \cdot (L - C) < -\vec{\pi}_a \cdot (C - W) \quad (13)$$

Example 1: Resume

- Only one price for traded insurance contracts.
- A continuum of equilibrium allocations with positive profits for the active intermediaries and underinsurance.
- Different equilibrium allocations correspond to a different latent allocation and to a different price for the latent contracts.

Relationship with Menu Games

A simple Menu Game: there are only TWO competitors (principals). Each of them can offer any combination of single offers (i.e. any subset of feasible contracts). The agent chooses one item from every principal's menu and takes her effort.

Result:

The equilibrium allocations $C \notin \mathcal{G}$ of our Example 1 can be supported as a symmetric equilibrium of a menu game with two principals.

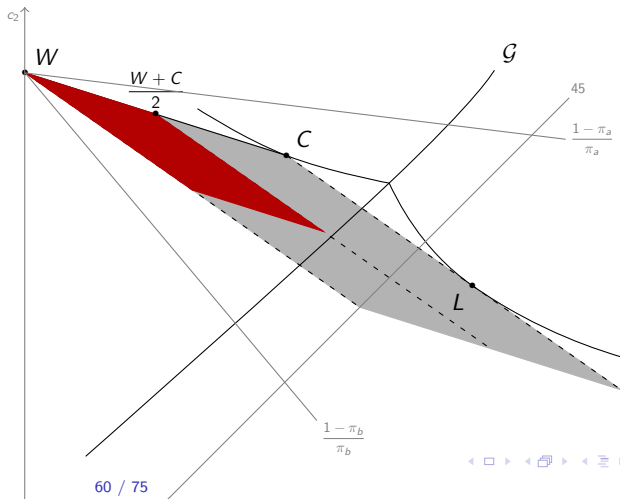
This could provide a rationale for latent contracts.

The Symmetric Equilibrium in Menus

Each of the two principals offer

$$\mathcal{M}^i = \left\{ \mu_i \left(\frac{W+C}{2} \right) + \lambda_i (L-C) \right\} \quad i \in \{1, 2\}.$$

The agent chooses $\mu_i = 1$ and $\lambda_i = 0$ in the menu of every principal.



Example 2: Two Active Intermediaries

We take again : $u(c) = c^\gamma$, $\gamma \in (0, 1)$.

An equilibrium with two active and two latent intermediaries:

- Intermediary 1 and 2 offer the positive insurance contracts d^1 and d^2 ,
- Intermediary 3 and 4 offer the same latent contract $\tilde{d}$,
- All other intermediaries do not make any offer $d^5 = \dots = d^J = (0, 0)$,
- The agent fully accepts the offer of intermediary 1 and 2, rejects all the others and chooses $e = a$.



Example 2: Resume

- Two prices for traded insurance contracts.
- The two financiers cannot be replaced by a single one offering the contract from W to C (there is not enough latent insurance in the market).
- Multiple financing is more likely to happen if the endowment is "far" from the 45-degree line.
- The equilibrium allocation C can be implemented as a non-symmetric equilibrium in a Menu Game.

Welfare: Constrained Efficiency

Benchmark: the Second Best. The set of feasible allocations for the planner must satisfy:

Aggregate Feasibility + Incentive Compatibility on the agent's decisions.

The planner directly controls (or can contract upon) the offers made by intermediaries. This guarantees a control on the consumption allocations, given the agent's optimal behavior.

The planner can always implement the desired allocation on the frontier $\mathcal{G}$ through outcome-contingent transfers. In particular, the (high effort) equilibrium allocation of the exclusive game can be implemented by having only one intermediary active.

Non-Exclusivity, which concept of Efficiency?

Under non-exclusivity, the planner loses control on the intermediaries' decisions, hence (indirectly) on the total insurance available to the agent.

Also, he does not control e nor λ_i .

The optimal allocation must be robust to side-contracting: (subsequent) strategic behaviors of intermediaries must be taken into account. The

only instrument is given by transfers to all players. Transfers can be state-contingent but NOT outcome-contingent. In particular, we consider the case of public insurance to the agent and fixed rewards to intermediaries.

Feasibility: No Side Contracting for Intermediaries

The planner's set of feasible allocations must satisfy:

- Aggregate Feasibility,
- Agent's Incentive Compatibility:

$$(e, \{\lambda_i\}_{i=1}^J) = \arg \max_{e', \{\lambda'_i\}_{i=1}^J} (1 - \pi_{e'})u(w_1 + \sum_{i=1}^J \lambda'_i d_1^i) + \pi_{e'} u(w_2 + \sum_{i=1}^J \lambda'_i d_2^i) - e'$$

- Robustness of the optimal allocation: (d^i, d^{-i}) has to be such that:

$$V^i(d^i, e(d^i, d^{-i}, W), \lambda_i(d^i, d^{-i}, W)) \geq -\tilde{\lambda}_i[(1 - \pi_{\tilde{e}})\tilde{d}_1^i + \pi_{\tilde{e}}\tilde{d}_2^i] \quad \forall i$$

High Effort Equilibrium and the Optimum: On Example 1

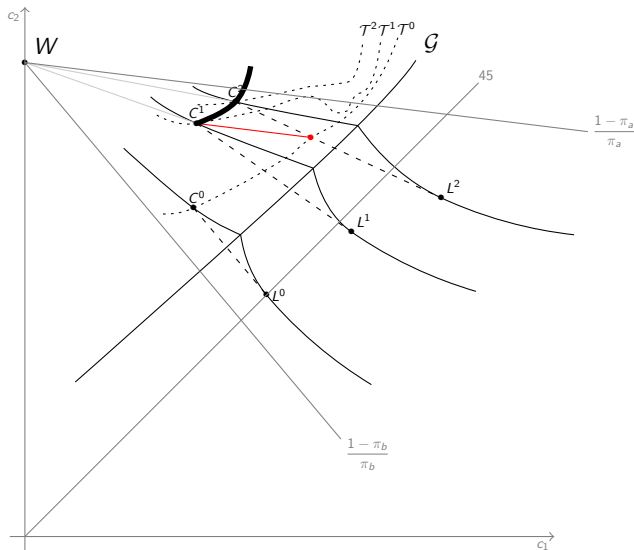
Fix any price of latent contracts $\tau < \frac{1-\pi_b}{\pi_b}$. At equilibrium, it must be:

$$U(C) = U(L), \quad (14)$$

$$\tau c_1 + c_2 = \tau l_1 + l_2, \quad \text{with } \tau = \frac{1 - \pi_b}{\pi_b} \frac{u'(l_1)}{u'(l_2)}. \quad (15)$$

- $\mathcal{T}$ is the curve that results from (14) and (15).
- $\mathcal{T}^O$ is the $\mathcal{T}$ curve corresponding to $\tau = \frac{1-\pi_b}{\pi_b}$.

All curves $\mathcal{T}$ are at the left of $\mathcal{T}^0$: $\tau^2 < \tau^1 < \frac{1 - \pi_b}{\pi_b}$



Third Best Efficiency and Prices of Latent Contracts

Result 1:

$\mathcal{T}^0$ is the frontier of the set of feasible allocations in the Third Best problem. Equilibrium allocations in the north-west of $\mathcal{T}^0$ fail to be third-best efficient.

Result 2:

In the case $u(c) = c^\gamma$ there is no equilibrium allocation associated to a $\tau = \frac{1-\pi_b}{\pi_b}$.

Public Insurance: Implementing an Allocation on T^0

Take any $C^0 = (c_1^0, c_2^0) \in \mathcal{T}^0$. The corresponding $L^0 = (l^0, l^0) \in \mathcal{B}$ will be such that:

$$l^0 = (1 - \pi_b)c_1 + \pi_b c_2$$

- The planner proposes to the agent the non-linear menu (public insurance):

$$\mathcal{M}^0 = (d^0 + \lambda \tilde{d}^0)$$

where $d^0 = C^0 - W$ and it is offered at a price $p^0 > \frac{1-\pi_a}{\pi_a}$,

$\tilde{d}^0 = L^0 - C^0$ and it is offered at a price $p^L = \frac{1-\pi_b}{\pi_b}$, and $\lambda \in [0, 1]$.

- Insurance companies propose the null contract, $d^1 = \dots = d^J = (0, 0)$.

Welfare Results

Proposition W1 When the planner provides public insurance by means of the piece-wise linear menu $\mathcal{M}^0$, the agent has the incentive to buy the amount d^0 of public insurance.

Also, it is a best reply for every private insurer to offer the null insurance contract.

For each positive-profit equilibrium allocation $C \in \mathcal{A}$ with latent insurance $L \in \mathcal{B}$ satisfying (14), (15), there exists a system of transfers and of public insurance that Pareto-dominates it.

Intuition

- $U(C^0) = U(L^0)$: the agent is indifferent between C^0 and L^0 .
- Private intermediaries cannot profitably deviate by reducing the agent's utility.
- A lower quantity and a higher price of latent insurance relative to the equilibrium level are sufficient to support the optimum.

Back to Game Theory:

Common Agency models of moral hazard

We consider a number of principals (indexed by $j \in N = \{1, \dots, K\}$) contracting with one agent.

The payoff to principal $k \in \{1, \dots, K\}$ and to the agent are given by the vNM utilities

$$v_k : \prod_{k \in \mathcal{K}} Y_k \times E \rightarrow \mathbb{R}_+,$$

$$u : \prod_{k \in \mathcal{K}} Y_k \times E \rightarrow \mathbb{R}_+.$$

Principals compete simultaneously offering menus. A menu $\hat{\gamma}_k$ is any subset of the set of feasible offers Y_k .

The agent chooses an item $y_k \in \hat{\gamma}_k$ from each principal j and a non contractible effort $e \in E$.

Standard analyses of competitive markets under asymmetric information assume that for each $k \in \mathcal{K}$ there exists a $V_k : Y_k \times E \rightarrow \mathbb{R}_+$ such that for all $(y_1, y_2, \dots, y_K) \in Y$, and for all $e \in E$,

$$v_k(y_1, y_2, \dots, y_K, e) = V_k(y_k, e)$$

Under moral hazard, the relevant payoffs for principals are those evaluated at the optimal level of effort.

That is, there are direct externalities.